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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (VDE) (CBCS) Degree Examinations

May-June 2025

22VDE201 – ANALOG VLSI DESIGN

Duration: 3 Hours

Max. Marks: 100

- Note:** i) Answer **Five full** questions choosing **One full** question from **each Unit**.
ii) Refer to the last page for the device SPICE parameters and equations.

Unit – I

Marks BT* CO* PO*

1. a) Develop the expression for MOSFET transconductance, g_m in terms drain current and overdrive voltage hence calculate g_m if $I_D = 1$ mA and overdrive voltage = 2V. Also plot g_m versus V_{DS} for the circuit shown in **Figure Q.1a**.

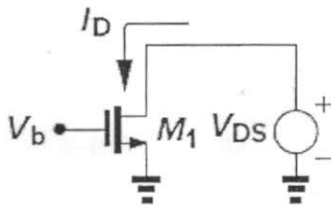


Figure Q.1a

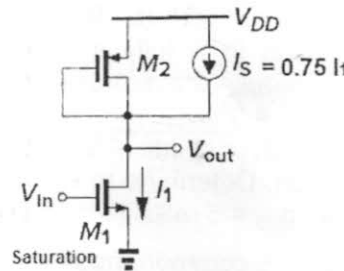


Figure Q.3a

- b) Write the complete small signal model of NMOSFET with $\gamma \neq 0$ and $\lambda \neq 0$. Also calculate the intrinsic gain of a NMOSFET with $\mu_n C_{OX} = 134 \mu A/V^2$, $(W/L) = 50/0.5$, $I_D = 0.5$ mA and $\lambda_{NMOS} = 0.1 V^{-1}$.
2. a) A MOSFET carries a drain current of 1 mA with $V_{DS} = 0.5$ V in saturation. Determine the change in I_D if V_{DS} rises to 1 V and $\lambda = 0.1 V^{-1}$. Also illustrate with characteristic plot of I_D versus V_{DS} for a NMOSFET for a given V_{GS} :
- With $\lambda = 0$ and $\lambda \neq 0$,
 - With $L = L_1$ and $L = 2L_1$
- b) Draw the small signal model of a NMOSFET with $\gamma = 0$ and $\lambda \neq 0$. Develop the expression for r_o . Determine the width of a NMOS current source with $I_D = 0.5$ mA operating with $V_{DS} = 0.4$ V. The minimum required r_o is 20 k Ω , $\lambda = 0.1 V^{-1}$ with $L = 0.5 \mu m$ and $V_{tn} = 0.7$ V.

10 L*3 1 3

10 L3 1 3

10 L3 1 3

10 L3 1 3

Unit – II

3. a) Determine the impedance of a diode connected NMOSFET. Hence determine the voltage gain of a common source amplifier with diode connected NMOS load using small signal circuit, with $\lambda = 0$ and $\gamma = 0$. Also determine the gain of CS stage show in **Figure Q.3a** if $(W/L)_1 = 100$ and $(W/L)_2 = 5$.
- b) Using small signal circuit, determine the maximum voltage gain of a source follower. In the source follower circuit shown in **Figure Q.3b**, the output resistance of M1 and M2 are $r_{o1} = 10$ k Ω and $r_{o2} = 15$ k Ω respectively. Determine the voltage gain if $g_{m1} = 0.5$ mho.

10 L3 2 3

10 L3 2 3

P.T.O

4. a) Assuming $\gamma = 0$ and $\lambda = 0$, derive the expression for the voltage gain of common source amplifier which employs negative feedback to linearize the gain. Also determine the gain by inspection of the circuit shown in **Figure Q.4a** if $\lambda = 0$, $R_D = 10\text{ k}\Omega$, $g_{m1} = 2\text{ mS}$, $g_{m2} = 1\text{ mS}$.

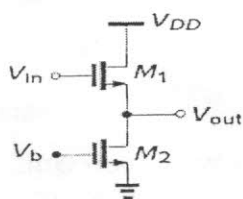


Figure Q.3b.

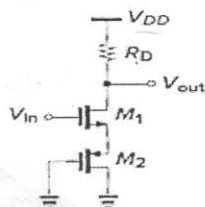


Figure Q.4a.

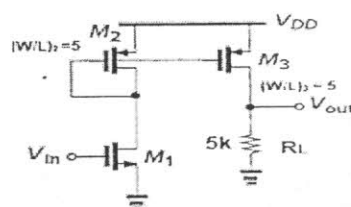


Figure Q.5b

- b) Design a source follower to drive a $50\ \Omega$ load with a voltage gain of 0.5 and a power budget of 10 mW. Assume $\mu_n C_{ox} = 100\ \mu\text{A/V}^2$, $V_{TH} = 0.5\text{ V}$, $\lambda = 0$, and $V_{DD} = 1.8\text{ V}$. Also recalculate the gain if the power budget is 15 mW.

Unit – III

5. a) Illustrate the virtual ground in a fully symmetric differential pair sensing differential input. Hence deduce the expression for the differential using half-circuit concept, assuming $\lambda \neq 0$.
b) Explain the basic current mirror circuit. Also explain development of cascode current mirror configuration. Determine the voltage gain of the circuit shown in **Figure Q.5b**. Assume $g_{m1} = 5\text{ mS}$.
6. a) Derive the expression for the common mode voltage gain of a differential pair with resistive load assuming circuit symmetry and non-ideal tail current source. Also calculate the differential gain of a differential pair with $V_{DD} = 3\text{ V}$, $R_{D1} = R_{D2} = 3.16\text{ k}\Omega$, $I_{Tail} = 1\text{ mA}$ with source resistance $R_{ss} = 500\ \Omega$. Given: $\mu_n C_{ox} = 50\ \mu\text{A/V}^2$, $(W/L)_{1,2} = 25/0.5$.
b) Construct a Gilbert cell and describe its application as an analog voltage multiplier.

Unit – IV

7. a) For a source follower, draw the high frequency equivalent circuit. Determine the transfer function and the most significant pole assuming $\gamma = 0$.
b) Determine the total output noise voltage in a common source stage with resistive load.
8. a) Explain input-referred noise voltage in a two-port network. Also determine the total input-referred noise voltage in a source follower with load implemented using a NMOSFET in saturation.
b) Draw the high frequency model of a common source stage. Determine the input pole and output pole hence write the expression for transfer function. State the errors in this estimation.

Unit – V

9. a) Draw the circuits of one-stage OPAMP and two-stage OPAMP. Briefly explain the same including the expression for the gain of each circuit.
b) Derive the mathematical model of a VCO. Also determine the output and its spectrum in a VCO with control voltage $V_{cont} = V_m \cos \omega_m t$.
10. a) Determine the gain and frequency of oscillation in each stage of a 3-stage ring oscillator. Also obtain the expression for the output.
b) Explain the application of PLL for frequency multiplication and frequency synthesis.

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})$$

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

$$g_m = \frac{2I_D}{V_{GS} - V_{TH}}$$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})(1 + \lambda V_{DS})$$

$$g_{mb} = g_m \frac{\gamma}{2\sqrt{2\Phi_F + V_{SB}}}$$

$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{2\Phi_F + V_{SB}} - \sqrt{|2\Phi_F|} \right)$$

$$\gamma = \sqrt{2q\epsilon_{si} N_{sub}} / C_{ox}$$

$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}$$

$$V_{TH} = \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_{ox}}$$

$$\Phi_F = (kT/q) \ln(N_{sub}/n_i)$$

$$Q_{dep} = \sqrt{4q\epsilon_{si} |\Phi_F| N_{sub}}$$

Level 1 SPICE models for NMOS and PMOS devices.

NMOS Model

LEVEL = 1	VTO = 0.7	GAMMA = 0.45	PHI = 0.9
NSUB = 9e+14	LD = 0.08e-6	UO = 350	LAMBDA = 0.1
TOX = 9e-9	PB = 0.9	CJ = 0.56e-3	CJSW = 0.35e-11
MJ = 0.45	MJSW = 0.2	CGDO = 0.4e-9	JS = 1.0e-8

PMOS Model

LEVEL = 1	VTO = -0.8	GAMMA = 0.4	PHI = 0.8
NSUB = 5e+14	LD = 0.09e-6	UO = 100	LAMBDA = 0.2
TOX = 9e-9	PB = 0.9	CJ = 0.94e-3	CJSW = 0.32e-11
MJ = 0.5	MJSW = 0.3	CGDO = 0.3e-9	JS = 0.5e-8
